

Supplementary Material: Decomposing the GP Advantage in Offline MBO

Anonymous submission

This supplement contains the proofs of Propositions 1–2, the full per-method result tables, the identifiability (matched-tuning) control, the β and ensemble-size ablations, the official-baseline cross-check, the significance details, and the complete experimental configuration. It is provided for reproducibility and is not part of the 7-page body.

1 Proofs

Notation. Objective $f : \mathcal{X} \rightarrow \mathbb{R}$; surrogate posterior mean μ and uncertainty $\sigma > 0$; LCB $L_\beta(x) = \mu(x) - \beta\sigma(x)$, $\beta \geq 0$. For a distribution Q on $\mathcal{X}$, L_β is a *valid* $(1-\delta)$ lower bound under Q if $\Pr_{x \sim Q}(f(x) \geq L_\beta(x)) \geq 1-\delta$.

Proposition 1 (Coverage of the premise is LCB validity). *For any Q, β, δ , $\Pr_{x \sim Q}(f(x) \geq L_\beta(x)) = \Pr_{x \sim Q}(\mu(x) - f(x) \leq \beta\sigma(x))$, so L_β is a valid $(1-\delta)$ lower bound under Q iff the premise $\{\mu - f \leq \beta\sigma\}$ has Q -probability at least $1-\delta$.*

Proof. Since $\sigma > 0$, pointwise $f(x) \geq \mu(x) - \beta\sigma(x) \iff \mu(x) - f(x) \leq \beta\sigma(x)$. The two events coincide as subsets of $\mathcal{X}$ and hence have equal probability under any Q . $\square$

Sanity checks. $\beta=0$ gives $\Pr(f \geq \mu) = \Pr(\mu - f \leq 0)$; $\beta \rightarrow \infty$ gives coverage $\rightarrow 1$. The identity is dimensionless and uses $\sigma > 0$ once.

Proposition 2 (Split-conformal repair; shift-limited transfer). *Let $\{(x_i, f(x_i))\}_{i=1}^n$ be exchangeable from P , $r_i = (\mu(x_i) - f(x_i))/\sigma(x_i)$ the signed one-sided nonconformity, and $\hat{q}$ the $\lceil (n+1)(1-\delta) \rceil$ -th smallest of $r_1, \dots, r_n$. Then for a fresh $x_{n+1} \sim P$ exchangeable with the calibration set, $\Pr(f(x_{n+1}) \geq \mu(x_{n+1}) - \hat{q}\sigma(x_{n+1})) \geq 1-\delta$. Under a shifted proposal $\Pi \neq P$ with density ratio $w = d\Pi/dP$, validity is restored by weighting the calibration quantile by w (weighted conformal).*

Proof. The normalized scores $r_1, \dots, r_{n+1}$ are exchangeable, so by the split-conformal rank argument $\Pr(r_{n+1} \leq \hat{q}) \geq 1-\delta$. The event $r_{n+1} \leq \hat{q}$ implies $\mu(x_{n+1}) - f(x_{n+1}) \leq \hat{q}\sigma(x_{n+1})$, i.e. $f(x_{n+1}) \geq \mu - \hat{q}\sigma$; monotonicity of probability gives the bound. The covariate-shift clause

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is the weighted-exchangeability extension of Tibshirani et al. (2019): replacing the empirical quantile of $\{r_i\}$ with its w -weighted analogue restores marginal validity under Π . At $\Pi = P$, $w \equiv 1$ recovers the unweighted bound. $\square$

2 Full Result Tables

Table 1 reports the full synthetic grid including the domain baselines (COMs, CbAS, gradient ascent); Table 2 the corresponding average ranks. The ensemble $\times$ gradient and gradient-ascent/COMs rows are the boundary-collapse cells discussed in the body.

3 Identifiability: Matched Tuning (GATE-1)

Exact and sparse GPs can spend per-run marginal-likelihood budget that ensembles cannot. The *matched* arm freezes GP hyperparameter fitting so every surrogate receives the same (zero) per-run tuning budget. The surrogate main effect on synthetic tasks is $\eta_{\text{sur}}^2=0.37$ unmatched and 0.28 matched—a 75% retention. The optimizer main effect stays small in both ($\eta_{\text{opt}}^2=0.01 \rightarrow 0.02$) and the interaction stays large ($\eta_{\text{inter}}^2=0.17 \rightarrow 0.12$). The GP advantage is therefore a surrogate-class property, not a per-run tuning artifact.

4 Mechanism Controls

The Section-5 mechanism rests on three controls, all on the synthetic tasks (task-normalized, averaged over optimizers).

Pessimism off ($\beta=0$). Re-running the full 3×3 grid at $\beta=0$ (no σ penalty) and reading both conditions on a single β -invariant normalizer, the GP-ensemble marginal gap is 0.319 [0.196, 0.460] at $\beta=0$ against 0.525 [0.406, 0.614] at $\beta=2$. Two things follow, and both are asserted. First, 61% of the advantage survives with σ removed entirely and the interval excludes zero, so the advantage rests on a substantial mean-quality base. Second, the paired increment is 0.203 [0.007, 0.396] with $p=0.020$, which *excludes* zero: pessimism is a significant amplifier of that base, not a by-stander. A normalization-free cross-check (per-task z -score by the task’s own pooled cell-mean standard deviation) gives 0.904 and 1.473 standard deviations respectively—the same 0.61 ratio—so the conclusion is not an artifact of min–max scaling. Per-task, the ensemble $\times$ gradient cell still collapses without pessimism (Griewank -42.7 , Rastrigin -9.5) while

Table 1: Full synthetic grid, 100th-percentile score (30 seeds), including domain baselines. **Bold** = best per task.

Method	Branin	Styblinski	Levy	Rosenbrock	Rastrigin	Ackley	Griewank
Ens+Grad	-9.27	6.37	-2.14	-0.28	-7.71	-3.66	-2592
Ens+Pert	-0.78	33.08	-0.40	-0.12	-8.44	-6.32	-395
Ens+CMA	-14.01	5.21	-3.19	-0.48	-10.81	-4.31	-2613
GP+Grad	-0.40	27.57	-0.05	-0.09	-4.83	-0.55	-0.94
GP+Pert	-0.40	36.15	-0.24	-0.08	-8.28	-6.28	-270
GP+CMA	-0.40	26.65	-0.05	-0.09	-5.06	-0.59	-1.00
SVGP+Grad	-0.45	11.83	-0.08	-0.04	-2.83	-0.69	-2.11
SVGP+Pert	-0.40	34.35	-0.25	-0.08	-8.27	-6.14	-275
SVGP+CMA	-0.53	11.60	-0.08	-0.04	-3.00	-0.73	-2.17
COMs	-9.73	8.50	-4.44	-1.31	-6.97	-4.49	-2078
CbAS	-5.64	33.85	-0.21	-0.05	-8.02	-5.10	-276
Grad. Asc.	-8.35	3.49	-2.43	-0.31	-9.08	-3.04	-2701

Table 2: Average rank over the 7 synthetic tasks (from Table 1; lower is better).

Method	Rank	Method	Rank
GP+Grad	2.57	GP+Pert	5.71
SVGP+Grad	3.29	SVGP+Pert	5.86
GP+CMA	3.57	CbAS	6.00
SVGP+CMA	4.29	Ens+Pert	8.14
Ens+Grad	8.57	COMs	9.43
Grad.Asc.	9.86	Ens+CMA	10.71

the GP does not (Griewank -0.96); these are the audited $\beta=0$ grid’s own cell means, on the same engine as the interval above.

Two cautions on these numbers. The per-task gaps are heterogeneous: on Branin the $\beta=0$ gap (0.664) *exceeds* its $\beta=2$ gap (0.468), while on Griewank-30D the $\beta=0$ gap is nearly absent (0.098); at $n=7$ this spread is descriptive only. And the figures above are *not* comparable to gaps computed with a normalizer refit separately at each β , which express the two conditions in different units; such figures may be read within a β condition but never as a cross- β ratio.

Matched subsample. Training the ensemble on the GP’s score-biased 800-point subsample rather than all data *widens* the gap to 0.76 (the ensemble marginal falls from 0.34 to 0.08): the GP wins with less data, so the biased subsample is not its edge. This control is computed on the legacy per-condition normalizer and is *not* on the same ruler as the $\beta=0$ figures above; only its direction should be read across the two.

Cross-proposal coverage. Table 3 measures each surrogate’s premise coverage in-distribution, on its own proposals, and on the other surrogate’s proposals. The ensemble premise holds in-distribution (0.73) and on the GP’s proposals (0.97) but collapses on its own (0.41); the GP premise holds on both (0.97). The ensemble is not miscalibrated in general—only where its own gradient optimizer drives it.

Table 3: Premise coverage at $\beta=2$ (synthetic mean), each surrogate on in-distribution points, its own proposals, and the other surrogate’s proposals.

Premise	in-dist	own prop.	other’s prop.
Ensemble	0.73	0.41	0.97
GP	0.98	0.97	0.93

5 Ablations

Pessimism strength β . Sweeping $\beta \in \{0, 0.5, 1, 2, 5\}$ (the per-task curves are Figure 4 of the main paper), increasing pessimism improves the final score on 6 of 7 synthetic tasks (median normalized slope $+0.19$); the exception is Ackley, where the uninformative σ makes the penalty point away from the optimum. That the benefit is distance-regularization rather than calibrated conservatism is confirmed by Figure 1: the per-task calibration quality ρ (rank correlation between σ and $|\mu - f|$) does not predict whether pessimism helps.

Ensemble size K . K is treated throughout as a confound to be controlled, never as a discovered mechanism. Sweeping $K \in \{2, 3, 5, 10\}$ on the audited engine (Figure 2), two facts are reported with equal weight. *Sensitivity:* η_{surr}^2 is K -dependent—0.326, 0.366, 0.408, 0.389—peaking at exactly the $K=5$ the field fixes by convention, so the headline magnitude sits at the effect’s maximum and we say so. *Non-reversal:* the ranking does *not* flip at $K=2$, where the ensemble marginal is 0.283 against the GP’s 0.767. Our pre-registered kill criterion for this sweep—that an ensemble marginal exceeding the GP’s at $K=2$ would overturn the headline—did not fire.

The monotonically decreasing task-normalized score reported in an earlier draft of this supplement (0.95 at $K=2$) *does not reproduce* on the audited engine, which gives 0.283 at that cell. That figure is withdrawn, and no claim in this paper rests on it.

6 Official-Baseline Cross-Check

We ran the official `design-baselines` COMs and CbAS and compare to our reimplementations on the normalized scale (Table 4). CbAS matches closely (TF-Bind-8

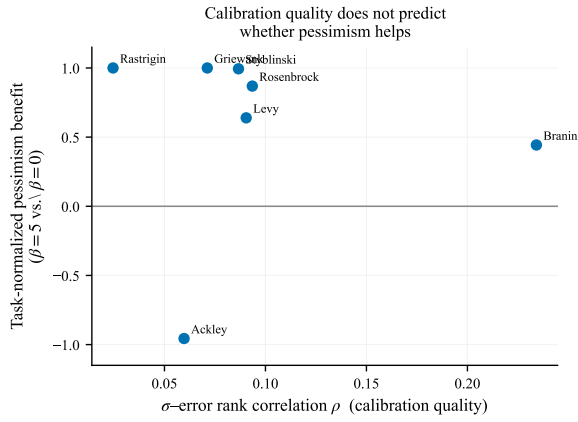


Figure 1: Calibration quality vs. pessimism benefit. Each point is a synthetic task; the σ -error rank correlation ρ (x-axis) does not predict the gain from pessimism (y-axis), consistent with a regularization rather than calibration effect.

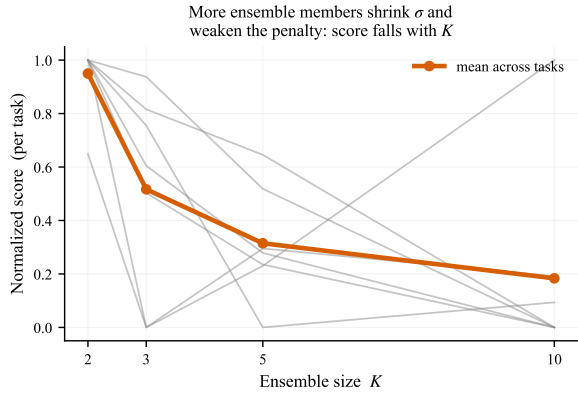


Figure 2: Ensemble-size ablation. Task-normalized score decreases with K .

$|\Delta|=0.004$); COMs diverges (TF-Bind-8 $|\Delta|=1.22$), reflecting a reduced-epoch official run and a different oracle variant. We therefore report both rather than a single number.

7 Per-Task Premise Coverage

Table 5 gives the per-task one-sided coverage of the pessimism premise at $\beta=2$, with the fitted split-conformal multiplier $\hat{q}$ and the repaired coverages. In-distribution coverage is moderate-to-high and on several real tasks above nominal (the ensemble under-predicts there, giving negative $\hat{q}$); OOD coverage is task-dependent, near zero exactly on the tasks where gradient ascent collapses. Conformal restores in-distribution coverage to its 0.90 target on every task but leaves OOD coverage erratic. GFP is degenerate—coverage is measured on relaxed logits and is a decode artifact rather than a calibration signal.

8 Significance Details

Over the 9 grid cells and 7 tasks the synthetic Friedman omnibus gives $p=6.1 \times 10^{-5}$ (Nemenyi critical difference

Table 4: Official vs. reimplemented baselines (normalized 100th-percentile).

Method	Task	Official	Ours	$ \Delta $
COMs	Superconductor	1.31	1.01	0.30
COMs	TF-Bind-8	0.99	2.21	1.22
COMs	GFP	−8.38	−9.20	0.82
CbAS	Superconductor	1.21	1.36	0.15
CbAS	TF-Bind-8	2.13	2.12	0.004
CbAS	GFP	2.20	1.85	0.35

Table 5: One-sided coverage of the premise $\mu - f \leq \beta\sigma$ (i.e. $f \geq \mu - \beta\sigma$) at $\beta=2$ (nominal 0.90). $c_{\text{in}}/c_{\text{ood}}$: coverage in-distribution / on proposals; $\hat{q}$: signed split-conformal multiplier (negative $\Rightarrow$ the mean already over-covers); c^{cf} : coverage after replacing β with $\hat{q}$.

Task	c_{in}	c_{ood}	$\hat{q}$	$c_{\text{in}}^{\text{cf}}$	$c_{\text{ood}}^{\text{cf}}$
Branin	0.71	0.42	6.2	0.91	0.84
Styblinski	0.64	0.00	7.5	0.90	0.00
Levy	0.68	0.11	6.0	0.90	0.28
Rosenbrock	0.86	0.64	2.5	0.90	0.66
Rastrigin	0.73	0.72	4.8	0.91	0.77
Ackley	0.92	1.00	1.8	0.89	1.00
Griewank	0.57	0.00	16.1	0.90	0.00
TF-Bind-8	0.92	0.71	1.7	0.90	0.70
TF-Bind-10	1.00	0.53	−1.6	0.89	0.45
Superconductor	1.00	0.01	−2.0	0.90	0.01
GFP	0.00	0.00	88.4	0.90	0.99
UTR	1.00	0.00	−2.5	0.91	0.00
Ant	0.94	0.00	1.2	0.90	0.00
D’Kitty	0.51	0.00	8.0	0.90	0.00

4.54; best cell GP+Grad, mean rank 2.29, bootstrap 95% CI [1.29, 3.57] over 10,000 task resamples). The Design-Bench omnibus over the same 9 cells gives $p=0.69$ (best cell GP+Pert, mean rank 3.57, CI [1.86, 5.71]; the entire grid within one critical difference). Adding the domain baselines (COMs, CbAS, gradient ascent) to the pool leaves both conclusions unchanged—the synthetic omnibus stays far below and the Design-Bench omnibus far above $\alpha=0.05$. Per-task Wilcoxon signed-rank tests with Holm correction and paired TOST equivalence tests are computed by the released `stats.py`; at the reported seed counts no Design-Bench pairwise comparison is significant after correction, and the omnibus null is not rejected.

Robustness to the random-forest oracle. Because four of the seven Design-Bench tasks use a substituted random-forest oracle (GFP, UTR, Ant, D’Kitty), we check that the null is not manufactured by RF-flattened surfaces. Restricted to the three tasks whose oracle is *not* a smoothing RF substitution—TF-Bind-8/10 (exact oracles) and Superconductor (native random-forest oracle)—the Friedman omnibus over the 9 cells is $p=0.93$, and the median per-task method spread on the substituted tasks is *no smaller* than on these three—0.39 normalized over the three substituted tasks other than the degenerate GFP, 0.51 over all four, against 0.34 here—so the

substitution did not compress the score range. The synthetic-to-real collapse of method separation is therefore a property of the real tasks, not an artifact of the oracle substitution.

9 Experimental Configuration

Surrogates. Ensemble: $K=5$ MLPs, two hidden layers of width 96, ReLU, MSE, 35 epochs, Adam (lr 3×10^{-3} , weight decay 10^{-4}), batch 256, per-member seed offset. Exact GP: differentiable single-task GP, ARD Matérn-5/2, marginal-likelihood fit (frozen under matched tuning), score-biased subsample to $N_{\max}=800$. SVGP: 128 inducing points, ARD Matérn-5/2, 250 variational-ELBO steps (Adam lr 0.01), $N_{\max}=2000$.

Optimizers. Gradient: 100 Adam steps on x (lr 0.05), box-clipped. Perturbation: 5 rounds, $\sigma \in \{0.1, 0.05, 0.02\}$, elementwise accept-if-better. CMA-ES: initial $\sigma=0.2$, bounds $[0, 1]$, separable (diagonal) covariance when $d>500$.

Acquisition and protocol. LCB $\mu - \beta\sigma$, $\beta=2$. Candidate budget 128 (top-128 dataset points plus $\sigma=0.05$ perturbations as starts; top-128 by surrogate score evaluated by the oracle). Metrics: 100th- and 50th-percentile normalized scores. Seeds: 30 synthetic, 16 Design-Bench, 10 for the β/K /calibration sweeps.

Tasks. Synthetic: Branin-2D ($N=2000$), Styblinski-5D (3000), Levy-8D (4000), Rosenbrock-10D (5000), Rastrigin-15D (5000), Ackley-20D (5000), Griewank-30D (8000); dataset drawn once at seed 0 and fixed across seeds. Design-Bench: TF-Bind-8/10 (exact oracles), Superconductor, GFP, UTR, Ant, D’Kitty (random-forest oracles); discrete designs relaxed to per-position class logits, decoded by argmax; inputs and scores min-max normalized; raw rows score-biased-subsampled to 8000 before encoding.

Calibration measurement. One-sided premise coverage $\widehat{\Pr}(\mu - f \leq \beta\sigma)$ (the LCB lower-bound event $f \geq \mu - \beta\sigma$) estimated in-distribution on uniform test points and out-of-distribution on the gradient-ascent proposals, at $\beta \in \{0.5, 1, 2, 5\}$. Split-conformal multiplier $\hat{q}$ from the signed nonconformity $(\mu - f)/\sigma$ with a relative σ floor (5% of the calibration-fold mean), fit at $\delta=0.1$ on a held-out fold; $\rho_{\text{err}} = \text{Spearman}(\sigma, |\mu - f|)$.

References

Tibshirani, R. J.; Foygel Barber, R.; Candes, E. J.; and Ramdas, A. 2019. Conformal Prediction Under Covariate Shift. In *Advances in Neural Information Processing Systems*.